TrailSense Multilateration Library: CLI Reference

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TrailSense Multilateration Library: CLI Reference

The trailsense-triangulate command-line interface contract: input JSON, output JSON, exit status, and diagnostics. Part of the TrailSense Multilateration Library documentation set.

CLI reference: trailsense-triangulate

The stdin/stdout/exit contract of the `trailsense-triangulate` command.

Synopsis

```
trailsense-triangulate < input.json > output.json
```

One JSON request from stdin, one JSON response to stdout. No arguments, options, or environment variables. Reads stdin to EOF before processing.

Input

A single JSON object with two required array keys, `anchors` and `obs`. Unknown top-level keys skipped. Trailing data after the top-level object is an error.

```
{
  "anchors": [ ... ],
  "obs":     [ ... ]
}
```

ANCHORS[]

Each element a surveyed sensor-node position; object non-empty.

Field	Type	Required	Default	Units / notes
`node_id`	integer	yes	none	must fit int32
`lat`	number	yes	none	degrees
`lon`	number	yes	none	degrees
`enrolled`	boolean	no	`true`	`false` excludes this anchor
`path_loss_a_cdb`	integer	no	`0`	centi-dB; `<= 0` means “use band default”
`path_loss_n_milli`	integer	no	`0`	milli; `<= 0` means “use band default”

OBS[]

Each element one node’s sighting of one device; object non-empty. All five fields required.

Field	Type	Required	Units / notes
`node_id`	integer	yes	must fit int32
`mac_hash`	string	yes	exactly 8 hex chars (4 bytes); case-insensitive on input
`band`	integer	yes	wire range `0..3` accepted; `0`=wifi, `1`=ble, `2`=cell; band `3` is parsed but rejected at solve time
`rssi`	integer	yes	dBm; must fit int8 (`−128..127`)
`ts_ms`	integer	yes	must fit uint32 (`0..4294967295`)

MAC_HASH FORMAT

8 hex chars (lower or upper case) mapping to 4 bytes: `"deadbeef"` becomes `de ad be ef`. Any length other than 8, or any non-hex character, is an error.

PARSER LIMITS

At most 256 anchors and 8192 observations per request; exceeding either is a fatal input error (nonzero exit). JSON nesting limited to 64 levels. Distinct from the core solver caps under STDERR below.

Output

On success, a single JSON object with one key, `positions`, an array. Empty when no group resolved: `{"positions":[]}`. Each element:

Field	Type	Units / notes
`mac_hash`	string	8 lowercase hex chars
`band`	integer	`0`=wifi, `1`=ble, `2`=cell
`signal_type`	string	`"wifi"`, `"bluetooth"`, or `"cellular"` (band 0/1/2)
`fingerprint_hash`	string	band prefix (`w_` / `b_` / `c_`) + the 8 hex `mac_hash`, e.g. `"w_deadbeef"`
`lat`	number	degrees, 9 decimal places
`lon`	number	degrees, 9 decimal places
`accuracy_m`	number	weighted CEP, meters; 2-node fix equals `error_semi_major_m`
`measurement_count`	integer	participating anchors (not post-outlier survivors)
`confidence`	integer	`40` (exactly 2 anchors), `70` (exactly 3), `85` (4 or more)
`peak_rssi`	integer	strongest participating rssi, dBm
`error_semi_major_m`	number	semi-major axis, meters
`error_semi_minor_m`	number	semi-minor axis, meters
`error_major_axis_deg`	number	bearing of semi-major axis (0=North, clockwise, folded into `[0,180)`)

`signal_type` and `fingerprint_hash` are CLI-added, not in `ts_position_t`. All other fields correspond one-to-one with the struct.

Exit status

Code	Meaning
`0`	Success. JSON written to stdout. Includes the zero-positions case (`{"positions":[]}`) and the case where a cap warning was emitted on stderr.
non-zero	Input error. A diagnostic on stderr; nothing usable on stdout.

Input errors: stdin not readable, malformed JSON, missing required field, empty anchor or obs object, `mac_hash` not exactly 8 hex chars, non-finite number, `node_id` / `rssi` / `ts_ms` out of range, `band` outside `0..3`, more than 256 anchors or 8192 obs, or trailing data after the top-level object. Never emits a partial or guessed position on bad input.

Stderr

Diagnostics only, never JSON. Two purposes:

- **Fatal errors** (nonzero exit): one line prefixed `trailsense-triangulate: input error:` or `trailsense-triangulate: ts_triangulate failed`.
- **Cap warnings** (exit 0): the core silently drops input past its static caps; the CLI detects this, warns, but still emits the partial result and exits 0. Two caps trigger a warning:
  - more than `TS_MAX_GROUPS` (64) distinct `(mac_hash, band)` groups, and
  - more than `TS_MAX_NODES_PER_GROUP` (32) distinct nodes within one group.

  Nonempty stderr with exit 0 means “partial result, some groups/nodes dropped,” not failure. Distinguish by exit code, not by whether stderr is empty.

Relationship to the C structs

Field semantics match the C structs exactly: `anchors[]` maps to `ts_anchor_t`, `obs[]` to `ts_obs_t`, each `positions[]` element to `ts_position_t` (plus CLI-only `signal_type` and `fingerprint_hash`). For field-level semantics including the mixed 2-node error-ellipse meaning, see the C API Reference and Output Reference.